

SENTINEL AI — ALL-IN-ONE SYSTEM DIRECTORY & ROLE MANUAL

Master Technical Guide: Complete Function Directory, Inputs & Outputs, Point-Wise Role Capabilities & Side-by-Side Role Comparison Table

Document Title: Master Directory & Role Capabilities	Classification: RESTRICTED DFIR / SOC
Coverage: 100% Platform APIs & UI Functions	AI Correlation Engine: Google Gemini 1.5 Flash

1. System Architecture & Platform Directory

SENTINEL AI is a unified Digital Forensics & Incident Response (DFIR) and Security Operations Center (SOC) platform. The architecture is designed around live real-time analysis engines: raw RFC-822 email parsing, PCAP frame dissection, static PE/ELF entropy calculation, WHOIS lookups, and SIEM event parsing, all integrated live with Google Gemini AI.

The system strictly segregates capabilities into three roles: **SOC Analyst**, **Forensic Investigator**, and **System Administrator**. Each role operates in a clutter-free, role-gated workspace.

1.1 Database Storage & Live Data Persistence Architecture

- ■ **Embedded Primary Database (SQLite):** Powered by Python standard library sqlite3 using ACID-compliant `forensics_platform.db` with Write-Ahead Logging (`PRAGMA journal_mode=WAL;`) for high concurrency and zero crash data loss.
- ■ **Enterprise Auto-Fallback (MongoDB):** Automatically connects to MongoDB via `pymongo` if an active MongoDB instance is detected at `mongodb://localhost:27017`. If offline, seamlessly switches to embedded SQLite without requiring any user setup.
- ■ **Atomic JSON Backup (`forensics_storage.json`):** Performs atomic double-write snapshots to `forensics_storage.json` on every database write operation to provide human-readable data recovery.
- ■ ■ **Live Data Retention Across Updates:** Startup hooks verify database integrity without purging user data, ensuring 100% of case history, evidence, security audit logs, dashboard metrics, and analyst alert triage statuses survive code updates and server restarts.

2. What Can Be Achieved by Different Roles (Point-Wise Capabilities)

2.1 SOC Analyst Role (Primary Goal: Real-Time Alert Triage & Threat Containment)

- ■ **Rapid Alert Triage:** Evaluate incoming SIEM and EDR security alerts in < 30 seconds. Analysts can dismiss false positives, issue immediate IP containment actions, or escalate critical alerts directly into investigation cases.
- ■ **Custom Threat Detection (YARA & Sigma):** Craft, test, and validate YARA rules and Sigma log detection patterns against live payload text to detect obfuscated PowerShell, Win32 API calls, and C2 traffic.
- ■ **SIEM Log Stream Anomaly Extraction:** Parse raw multi-format log text streams (`.evtx``, Syslog, JSON) to automatically isolate brute-force bursts, shell execution events, and identify top IP talkers.
- ■ **External Attack Surface & CVE Auditing:** Perform external infrastructure health audits to discover open high-risk ports, SSL certificate status, and matching CVSS vulnerabilities on target assets.
- ■ **Instant SOAR Containment Playbooks:** Generate 1-click executable mitigation scripts across Linux (``iptables`` / BIND sinkholes) and Windows PowerShell firewalls to immediately block compromised hosts, IPs, or domains.

2.2 Forensic Investigator Role (Primary Goal: Incident Reconstruction & Case Building)

- ■ **Official Incident Case Management:** Create and maintain formal investigation cases, assign threat severity levels (CRITICAL / HIGH / MEDIUM / LOW), track operational status, and assign case owners.
- ■ **Multi-Vector Artifact Correlation:** Upload and attach email RFC-822 headers, live URL scans, network PCAP packet files, and malware binaries directly to a unified case timeline.
- ■■ **Chronological Attack Reconstruction:** Automatically synthesize multi-file evidence sources into a unified, step-by-step chronological attack timeline.
- ■ **Printable Executive PDF Reports:** Generate professional, printable forensic investigation PDF reports complete with executive summaries, evidence tables, IOC indicators, and remediation blueprints.
- ■ **AI Threat Correlation:** Leverage Google Gemini AI to analyze complex multi-vector evidence files and identify hidden cross-artifact relationships.

2.3 System Administrator Role (Primary Goal: Governance, RBAC & Platform Health)

- ■ **User & Role Management:** Dynamically switch user accounts between `Analyst`, `Investigator`, and `Admin` roles with immediate real-time role updates.
- ■ **AI Engine Key Management:** Configure and persist global Google Gemini API keys across local storage and database backend.
- ■ **Platform-Wide Telemetry & Audit Logs:** View global multi-user metrics, aggregate case totals, and monitor security audit logs across all platform operators.

3. Comprehensive Role Comparison Table

The following side-by-side comparison matrix contrasts the permissions, primary objectives, inputs, outputs, and access boundaries of all three platform roles:

Comparison Dimension	SOC Analyst (`analyst`)	Forensic Investigator (`investigator`)	System Admin (`admin`)
Primary Operational Goal	Rapid alert triage, threat hunting, log parsing & fast containment	Deep-dive case building, multi-evidence correlation & executive reporting	System governance, user role management, AI key config & audit logging
Key Workspace / Route	Analyst Workspace (`/analyst`)	Cases & Incidents (`/cases`)	Admin Control Panel (`/admin`)
Primary Input Data	Triage alerts, raw SIEM log text, YARA/Sigma rules, Target IPs/Domains	Case titles, RFC-822 emails, PCAP files, PE binaries, URL links	Gemini API keys, user role selections, audit log queries
Primary Output Data	Triage statuses, YARA match lines, log anomaly tables, SOAR block scripts	Unified attack timelines, case artifact links, Printable PDF reports	Updated user roles, active AI status, platform audit log stream
Case Management Access	■ Hidden & Restricted (Auto-escalation only)	■ FULL ACCESS (Create, Edit, Delete, Attach)	■ FULL ACCESS (Global view across all users)
Analyst Tools Access	■ FULL ACCESS (Triage, YARA, Logs, SOAR)	■ Hidden & Restricted	■ FULL ACCESS
PDF Report Generation	■ Hidden & Restricted	■ FULL ACCESS (Printable PDF case reports)	■ FULL ACCESS
Admin Control Access	■ Hidden & Restricted	■ Hidden & Restricted	■ FULL ACCESS (Manage users & API keys)

4. Master Platform Function Directory (All Functions & APIs)

Below is the complete function directory detailing inputs, outputs, API routes, and structured return data for all platform tools:

4.1 Email Forensics Engine

POST /api/evidence/parse-email	Route: /email-forensics Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Upload raw `.EML` or `.MSG` email file OR paste raw RFC-822 MIME headers.
INPUT EXAMPLE:	From: attacker@malicious-c2.xyz To: victim@company.com Subject: Urgent Payment Needed Received: from 185.220.101.5
OUTPUT GENERATED:	Extracts hop transit timeline, verifies SPF/DKIM/DMARC authentication, and hashes attachments.
OUTPUT FIELDS:	• hop_trace: Array of IP hops and server hostnames. • auth_results: SPF status, DKIM signature validity, DMARC alignment. • attachment_hashes: MD5 & SHA-256 checksums of extracted email attachments. • threat_level: CRITICAL / HIGH / MEDIUM / LOW rating.

4.2 URL Threat & Infrastructure Analysis Engine

POST /api/evidence/parse-url	Route: /url-forensics Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Type or paste any target URL, domain name, or IP address into search box.
INPUT EXAMPLE:	https://login.paypal-security-update.account-verify.xyz/login.php
OUTPUT GENERATED:	Live DNS resolution, WHOIS registrar queries, SSL certificate checks, and brand spoofing detection.
OUTPUT FIELDS:	• resolved_ip: Live server IP address, country, ISP. • domain_age_days: Days since domain WHOIS registration. • shannon_entropy: Randomness rating of URL path. • target_brand_spoof: Identified typosquatting brand target.

4.3 Network PCAP Packet Dissector

POST /api/evidence/parse-pcap	Route: /network-forensics Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Upload binary `.pcap` or `.pcapng` packet capture file (max 50MB).
INPUT EXAMPLE:	binary_capture.pcap (Ethernet/IP frame byte stream)
OUTPUT GENERATED:	Dissects binary frames, reconstructs TCP/UDP streams, and extracts DNS queries.
OUTPUT FIELDS:	• total_packets & total_bytes: Packet count & bandwidth volume. • protocol_breakdown: HTTP, HTTPS, DNS, SSH, FTP traffic ratios. • top_talkers: Source and Destination IPs ordered by packet count. • dns_queries: Array of resolved domain names.

4.4 Malware Static PE Analysis Engine

POST /api/malware/analyze-binary	Route: /malware-forensics Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Upload executable binary file (.exe, .dll, .bin, .elf, .apk, .pdf).
INPUT EXAMPLE:	update_svc.exe (binary Windows Portable Executable)
OUTPUT GENERATED:	Calculates section entropy, extracts Win32 API imports, and flags packing.
OUTPUT FIELDS:	• file_hashes: MD5 & SHA-256 binary checksums. • entropy_score: 0.00–8.00 rating (Entropy > 7.2 flags packing). • suspicious_api_calls: Flagged Win32 APIs (VirtualAllocEx, WriteProcessMemory). • malware_family_hint: Classification hint (Ransomware, Spyware, Trojan).

4.5 Multi-Feed Threat Intelligence Lookup

POST /api/threat-intel/enrich	Route: /threat-intelligence Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Enter any IP address, domain name, file hash (MD5/SHA256), or CVE ID.
INPUT EXAMPLE:	185.220.101.5 OR c5b3a...4f2e
OUTPUT GENERATED:	Queries global threat intelligence feeds for malicious reputation scores.
OUTPUT FIELDS:	• reputation_score: Malicious detection ratio. • known_campaigns: Associated APT groups or malware strains. • asn_owner: Autonomous System Number & hosting organization.

4.6 Analyst SOC Workspace Functions

POST /api/analyst/triage & /test-rule	Route: /analyst Roles: Analyst, Admin Only
INPUT FORMAT:	• Alert Action (FALSE_POSITIVE, CONTAIN, ESCALATE) • YARA / Sigma Rule Code & Test Payload Sample
INPUT EXAMPLE:	<pre>Rule Code: `rule Detect_Powershell { strings: \$s1 = "VirtualAllocEx" condition: \$s1 }` Payload: `powershell.exe -enc ... VirtualAllocEx`</pre>
OUTPUT GENERATED:	Updates alert status (or escalates to Case), and evaluates YARA/Sigma signature matching.
OUTPUT FIELDS:	• triage_status: New alert status (FALSE_POSITIVE, CONTAINED, ESCALATED_TO_CASE). • hunting_result: Match Status (MATCH_FOUND / NO_MATCH), match count, matched line snippets.

4.7 SOAR Playbook Generator

POST /api/analyst/playbooks/generate	Route: /analyst Roles: Analyst, Admin Only
INPUT FORMAT:	Select Action Type ('BLOCK_IP', 'BLOCK_DOMAIN', 'ISOLATE_HOST') and enter Target Value.
INPUT EXAMPLE:	Action: BLOCK_IP Target: 185.220.101.5
OUTPUT GENERATED:	Generates 1-click executable block scripts for Linux and Windows systems.
OUTPUT FIELDS:	• bash_payload: Executable Linux 'iptables' / BIND sinkhole commands. • powershell_payload: Executable Windows PowerShell firewall commands.

4.8 Case Management & Admin Governance Functions

POST /api/cases & PUT /api/admin/users/{id}/role	Route: /cases & /admin Roles: Investigator, Admin
INPUT FORMAT:	• Case Title, Description, Category, Priority Level • Target User ID & New Role ('analyst', 'investigator', 'admin')
INPUT EXAMPLE:	Case: Title='Mail Intrusion', Priority='CRITICAL' Admin Update: UserID='user@company.com', Role='analyst'
OUTPUT GENERATED:	Creates official investigation case, or dynamically updates user role permissions.
OUTPUT FIELDS:	• case_number: Unique Case ID (e.g., CASE-2026-9041). • user_role_update: Updated user permissions and role assignment.

5. Platform File Directory & Architecture Map

The platform backend and frontend code structure is mapped below for reference:

- **Backend Core**: backend/main.py (FastAPI app & CORS setup), backend/app/database.py (JSON / MongoDB store)
- **Authentication & RBAC**: backend/app/auth.py, backend/app/routers/auth_router.py
- **Analyst Workspace APIs**: backend/app/routers/analyst_router.py ('/api/analyst/alerts', '/test-rule', '/parse-logs', '/scan-attack-surface', '/playbooks/generate')
- **Case Management APIs**: backend/app/routers/cases_router.py ('/api/cases'), backend/app/routers/evidence_router.py
- **Forensics Engines**: backend/app/routers/malware_router.py, backend/app/routers/threat_intel_router.py, backend/app/routers/gemini_router.py
- **Frontend Views**: frontend/src/pages/AnalystPage.jsx, Cases.jsx, CaseDetails.jsx, AdminPanel.jsx, Dashboard.jsx
- **Navigation & RBAC Guards**: frontend/src/components/Sidebar.jsx, MobileBottomNav.jsx, Navbar.jsx, App.jsx